Aman Yadav

+91-9569491993 | amanstorm77@gmail.com | GitHub | LinkedIn | Portfolio

Summary

Mathematics & Computing undergraduate focused on backend systems and distributed task processing. Built asynchronous worker pipelines, relational database architectures, and REST APIs using Python, FastAPI, Django, PostgreSQL, Redis, Celery, and Docker.

Education

Central University of Karnataka Kalaburagi, Karnataka, India
Bachelor of Technology in Mathematics & Computing *2023 – 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Distributed Computing, Object-Oriented Programming, Linear Algebra, Probability & Statistics, Numerical Optimization.

Experience

Maruti Suzuki India Limited Gurugram, India
Engineering Intern (Applied ML & Telematics) – EN-DAX Dept *May 2026 – Jul 2026*

- Developed a 6-stage ingestion pipeline normalizing raw **CAN-bus** sensor streams (up to 1M+ rows) into a **40-feature taxonomy** with automated trip segmentation.
- Derived kinematic features using vectorized **Haversine distance** and speed differentials to detect hard braking and rapid acceleration, applying dual-stage **IQR and Z-score** outlier filtering.
- Delivered a **Streamlit** console for vehicle dynamics engineers to inspect sensor time series, trip heatmaps and kinematic distributions; learned to handle noisy real-world data and design validation with audit trails.

Projects

PingGuard – Distributed HTTP Uptime Monitoring Engine GitHub Repo
FastAPI, PostgreSQL, Redis, Celery, SQLAlchemy, Docker

- Decoupled the **FastAPI** control plane from outbound network I/O via **Celery** worker fleets on Redis, applying exponential backoff to recover from transient timeouts.
- Engineered DB-driven dispatch via PostgreSQL **row-level locking** (`SELECT ... FOR UPDATE SKIP LOCKED`) with recovery transactions to prevent duplicate or lost tasks.
- Hardened probing against **SSRF** and TOCTOU **DNS rebinding** by enforcing asynchronous pre-flight DNS checks against private CIDR blocklists and directly pinning target IPs.

EduPulse – Academic Forecasting & Early-Warning Platform GitHub Repo
Django, PostgreSQL, Redis, Celery, Scikit-Learn, Docker

- Built a modular REST API backend in Django with role-based access control (**RBAC**), **JWT** authentication, and transactional persistence in PostgreSQL.
- Configured an isolated dual-Redis architecture separating caching from Celery task queues, applying **generational cache invalidation** to maintain response times **under 40 ms**.
- Built batch ML inference pipelines using **scikit-learn** with `GroupedKFold` cross-validation and demographic quarantine to keep demographic attributes out of model features; validated via **220+ automated tests**.

EV-Lifespan – EV Battery Degradation & RUL Platform GitHub Repo
FastAPI, PyTorch, Docker, Scikit-Learn, REST APIs

- Built a low-latency **FastAPI** inference service serving **PyTorch** models at under 15 ms latency, validating incoming payloads with a 19.40-cycle RMSE on NASA datasets.
- Packaged the service using a **multi-stage Docker build** with automated container health checks, structured error logging, and strict REST API contract validation.

Technical Skills

Languages: Python, SQL, C++, Java, MATLAB

Backend & Infrastructure: FastAPI, Django, Celery, REST APIs, Docker, JWT/RBAC

Databases: PostgreSQL, Redis, SQLAlchemy

ML & Data: PyTorch, scikit-learn, Pandas, NumPy, Streamlit

Tools: Git, GitHub Actions, pytest, Docker Compose, Postman